```c
#include <stdio.h>
int main(){
    int numberOfProcesses;
    printf("Enter number of processes: ");
    scanf("%d", &numberOfProcesses);

    int processId[10];
    int burstTime[10];
    int priority[10];

    // Input process details
    for(int i = 0; i < numberOfProcesses; i++)    {
        processId[i] = i + 1;

        printf("Enter burst time of P%d: ", processId[i]);
        scanf("%d", &burstTime[i]);

        printf("Enter priority of P%d: ", processId[i]);
        scanf("%d", &priority[i]);
    }

    // Sort by priority (lower value = higher priority)
    for(int i = 0; i < numberOfProcesses - 1; i++)
    {
        for(int j = i + 1; j < numberOfProcesses; j++)
        {
            if(priority[j] < priority[i])
            {
                int temp;

                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;

                temp = burstTime[i];
                burstTime[i] = burstTime[j];
                burstTime[j] = temp;

                temp = processId[i];
                processId[i] = processId[j];
                processId[j] = temp;
            }
        }
    }

    int waitingTime[10];
    int turnaroundTime[10];
    int totalWaitingTime = 0;
    int totalTurnaroundTime = 0;

    // Calculate waiting and turnaround time
    for(int i = 0; i < numberOfProcesses; i++)    {
        if(i == 0)
            waitingTime[i] = 0;
        else
            waitingTime[i] = waitingTime[i - 1] + burstTime[i - 1];

        turnaroundTime[i] = waitingTime[i] + burstTime[i];
        totalWaitingTime += waitingTime[i];
        totalTurnaroundTime += turnaroundTime[i];
    }

    // Output
    printf("\nPID\tBT\tPR\tWT\tTAT\n");
    for(int i = 0; i < numberOfProcesses; i++)    {
        printf("P%d\t%d\t%d\t%d\t%d\n",
               processId[i],
               burstTime[i],
               priority[i],
               waitingTime[i],
               turnaroundTime[i]);
    }
    printf("Average Waiting Time = %.2f\n",           (float)totalWaitingTime / numberOfProcesses);
    printf("Average Turnaround Time = %.2f\n",           (float)totalTurnaroundTime / numberOfProcesses);
}
```